

Characterization and modeling of the persistent high-latitude ionosphere response to the solar wind

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Abstract

The ionosphere-thermosphere system is strongly influenced by solar EUV and solar wind energy input. The solar wind drives a persistent response of the high-latitude ionosphere under winter nighttime conditions. The drivers of this response were quantified using a time-lagged cross-correlation analysis of 26-years of solar wind and ionosphere observations [1]. Modeling of the ionosphere response and its spatial variability in the high-latitude region remains a challenge associated with the magnetosphere forcing and high-latitude electrodynamics. To evaluate the model capabilities on reproducing the high-latitude ionosphere response to solar wind, we perform multiple simulations for a 90-day winter period using the NCAR Thermosphere-Ionosphere-Electrodynamics General Circulation Model (TIE-GCM), driven by different empirical and data-assimilated convection forcing models.

Introduction

The persistent ionosphere response to solar wind forcing during high-latitude winter night conditions is characterized by the variations of ionospheric total electron content (TEC) with a delay of ≈ 120 minutes, which was found to be driven mainly from the F-region plasma convection processes (≈ 90 -135 mins), with the merged effect of magnetospheric loading-unloading processes and auroral particle precipitation (≈ 45 -90 mins) [1].

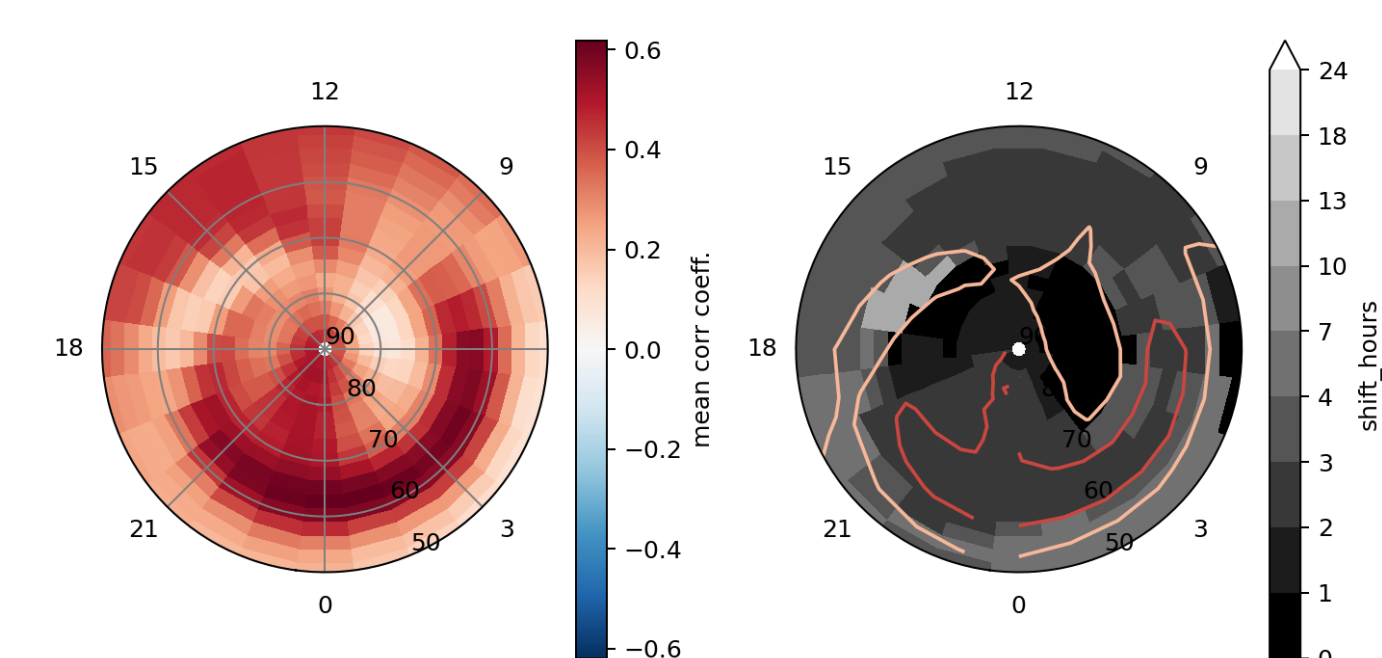


Fig.1: Lagged cross-correlation (left) and the corresponding time lag (right) between IGS TEC and OMNI merging electric field, E_m , in the selected winter period of 01.01.2013 $\pm$ 45 days, averaged over all UT [1].

Physics-based numerical models include mechanisms such as impact from below (atmospheric forcing) and above (magnetospheric forcing) that cause variations in the ionosphere dynamics and energetics, and thus they are an important tool for understanding the ionosphere response to solar wind. The TIE-GCM is a global physics-based numerical model of the coupled ionosphere-thermosphere system that solves the fully coupled nonlinear equations describing the neutral and ionized atmosphere self-consistently [3,4].

Model Setup

To evaluate the model capabilities at high-latitudes, multiple TIE-GCM v2.0 runs during the winter of 15 November 2012 - 15 February 2013 ($Kp \leq 4.67$) with varying electric potential model choices: AMGeO (data-assimilated), Weimer and Heelis (empirical) [5,6,7] were performed. The auroral particle precipitation is represented by the auroral scheme of [8] with modifications of [9] in all runs. At the lower boundary of 97 km altitude, atmospheric dynamics are driven by the Global Scale Wave Model (GSWM) [10]. To evaluate the quality of the model performance, we use statistical parameters including relative bias (RB) calculated by using the TIE-GCM TEC and IGS/UPC TEC data.

Table 1: TIE-GCM run descriptions for the selected winter period of 01.01.2013 $\pm$ 45 days

Run	Date	Convection	Spatial resolution	Temporal resolution
1	15/11/2012-15/02/2013	AMGeO	$2.5^\circ \times 2.5^\circ$	60 min
2	15/11/2012-15/02/2013	Weimer	$2.5^\circ \times 2.5^\circ$	20 min
3	15/11/2012-15/02/2013	Heelis	$2.5^\circ \times 2.5^\circ$	20 min
4	15/11/2012-15/02/2013	IGS /UPC	$2.5^\circ \times 5^\circ$	15 min

Model Results

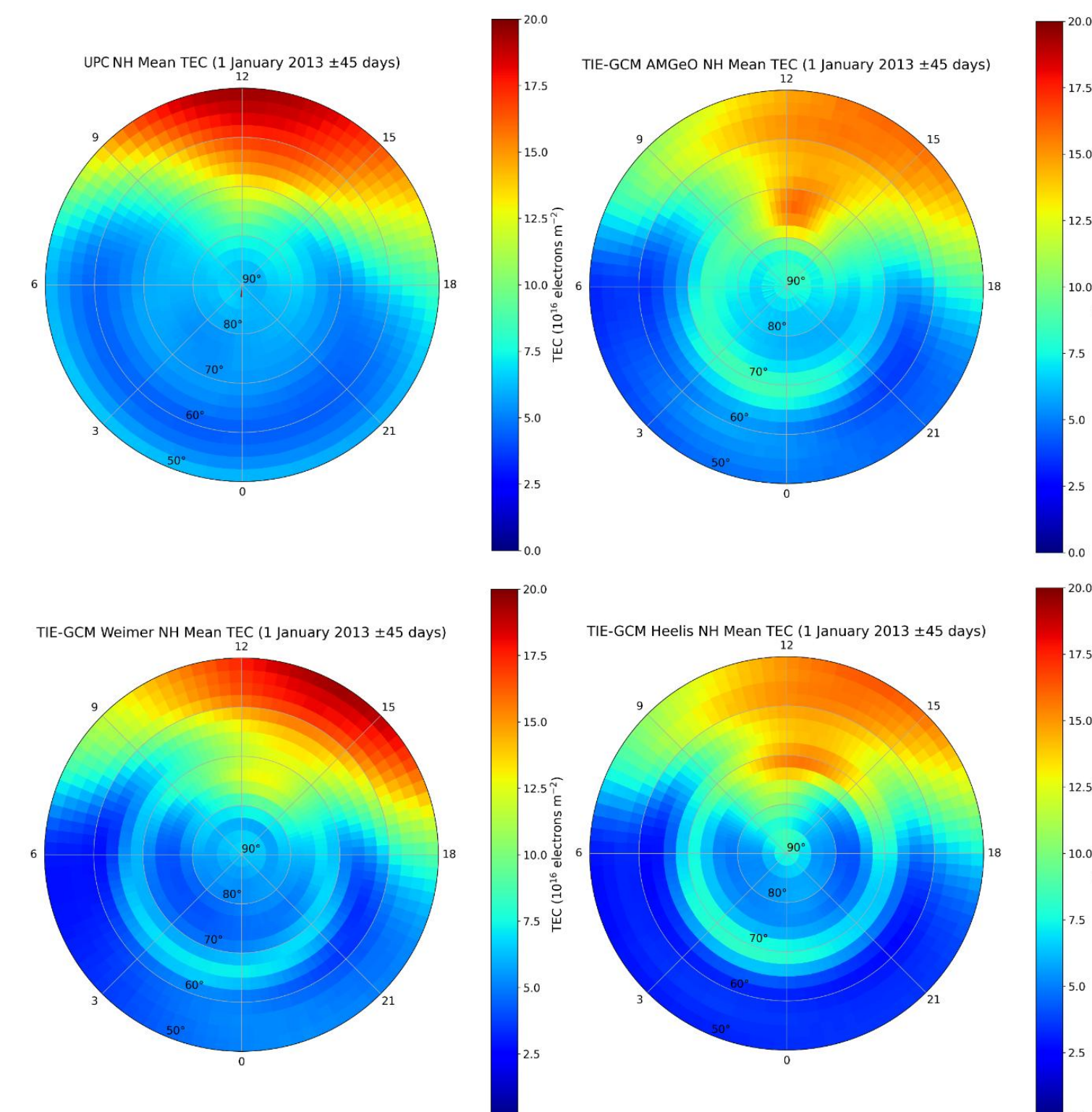


Fig.2: Mean TEC (TECU) averaged on the 90-days northern hemisphere winter period of 01.01.2013 $\pm$ 45 between 50° - 90° MLATs binned for scale of 0.25. Top left: IGS/UPC, top right: TIE-GCM AMGeO, bottom left: TIE-GCM Weimer, bottom right: TIE-GCM Heelis.

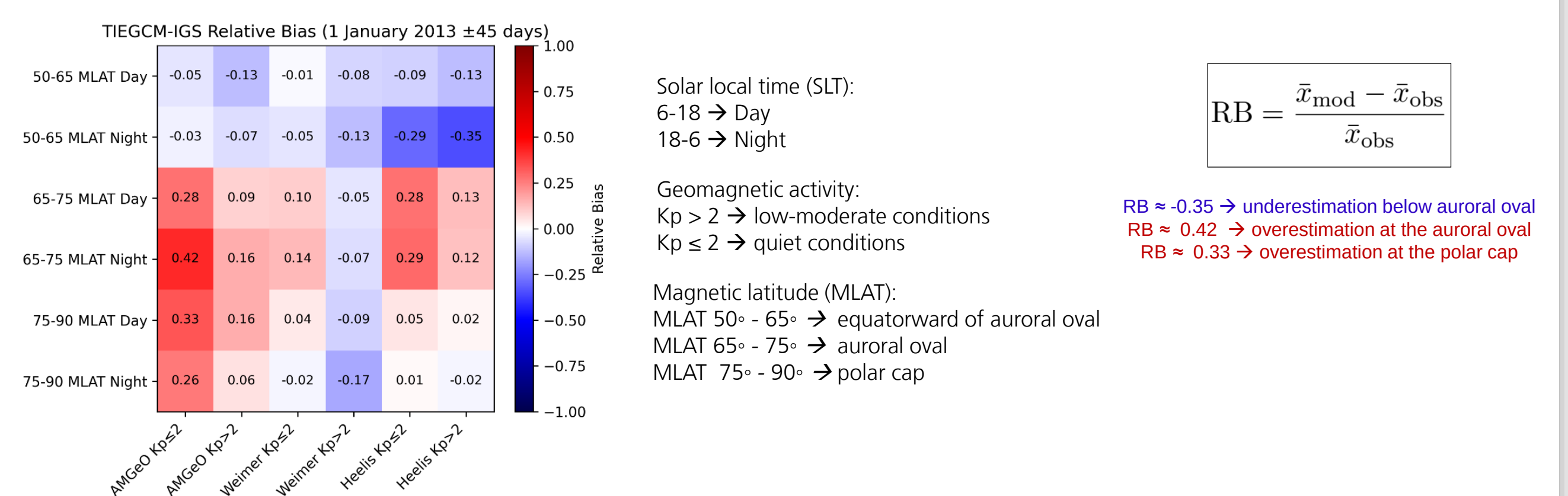


Fig.3: Relative TEC difference bias (RB) of different TIE-GCM runs with three convection forcing and IGS/UPC TEC, averaged for the winter period of 01.01.2013 $\pm$ 45.

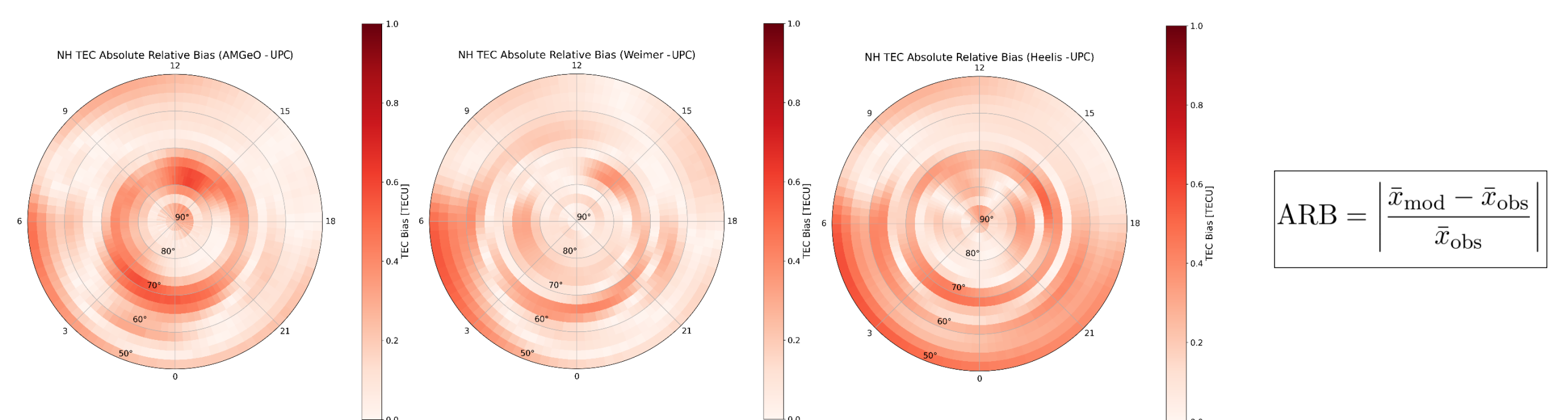


Fig.4: Absolute relative TEC difference bias (ARB) of different TIE-GCM runs with three convection forcing and IGS/UPC TEC, averaged for the winter period of 01.01.2013 $\pm$ 45 between 50°-90° MLATs binned for scale of 0.25. Left: TIE-GCM AMGeO - IGS/UPC, middle: TIE-GCM Weimer - IGS/UPC, right: TIE-GCM Heelis - IGS/UPC.

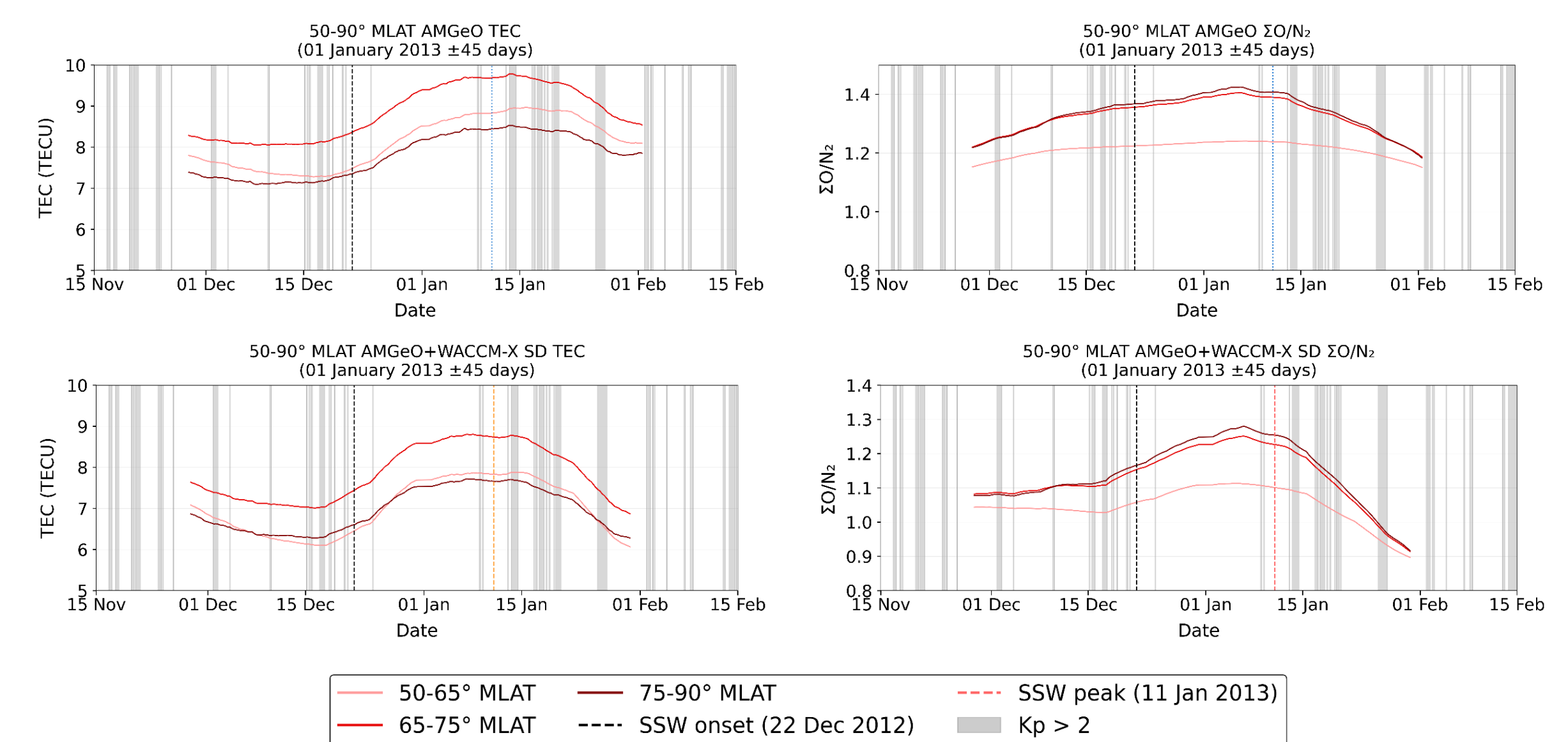


Fig.5: The effect of lower atmospheric forcing on the data-assimilated TIE-GCM + AMGeO TEC and O/N2 for the winter period of 01.01.2013 $\pm$ 45 between 50°-90° MLATs. Top: TIE-GCM + AMGeO. Bottom: TIE-GCM + AMGeO + WACCM-X with specified dynamics (WACCM-X SD).

Summary

- TIE-GCM simulations under geomagnetically quiet-moderate winter conditions are not fully capable of representing the observed high latitude ionosphere response due to solar wind, independently from the choice of convection forcing model (AMGeO, Heelis, Weimer).
- The overestimation of TEC mainly in the auroral oval (65° - 75° MLAT) and polar cap (75° - 90° MLAT), alongside with the underestimation of TEC below the auroral oval latitudes (50° - 65° MLAT) is observed.
- The model-observation difference was attributed to the auroral parametrization, and the thermosphere composition and circulation processes.
- The thermosphere-ionosphere variations due to the major stratospheric warming (SSW) event taking place in the 2012-2013 winter period, are not observed in the model runs, except if the lower boundary conditions are specified (WACCM-X SD).

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